

Sri Lanka Institute of Information Technology



Web-based food ordering system

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Software Engineering | SE2030

IT Number	Name
IT25101692	Wijesingha A.S.
IT25102506	Isurujith M. U.
IT25103590	Sajeenth T.
IT25101506	De Silva K.T.D.A.
IT25103393	Diyana H.E.S
IT25100590	Rishan J R

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2. Introduction

Project Overview

The rapid advancement of information technology has transformed the way restaurants manage their daily operations and interact with customers. Many customers now expect the convenience of browsing menus, placing orders online, and choosing between pickup or delivery services. However, restaurants that rely on traditional ordering methods often face challenges such as inefficient order handling, inconsistent menu management across branches, and difficulties in coordinating deliveries and customer feedback.

To address these challenges, this project proposes **Spice Avenue**, a web-based food ordering system developed for a restaurant company operating multiple branches. The system provides a centralized platform that enables customers to place pickup and delivery orders while allowing restaurant staff to efficiently manage branches, menus, customer orders, deliveries, and customer feedback. By digitalizing these business processes, the system aims to improve operational efficiency, enhance customer convenience, and support better decision-making across restaurant branches.

Project Objectives

The primary objective of this project is to develop a centralized web-based food ordering system that improves restaurant operations and enhances the customer ordering experience.

The specific objectives of the project are:

- Develop a centralized food ordering system for a restaurant company with multiple branches.
 - Enable customers to browse menus and place pickup or delivery orders conveniently.
 - Improve the efficiency of order processing and restaurant operations.
 - Support effective management of branches, menus, deliveries, and customer feedback.
 - Provide customers with order tracking throughout the ordering process.
 - Improve customer satisfaction by providing a convenient, reliable, and user-friendly ordering platform.
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Target Users / Stakeholders

The proposed system will be used by **Customers, Branch Managers, Delivery Riders, Customer Service Supervisors, Operations Managers, and System Administrators**. Each user interacts with the system according to their assigned responsibilities, enabling efficient management of restaurant operations while providing customers with a convenient and reliable food ordering experience.

Project Scope

The project focuses on developing a web-based food ordering system that supports the complete food ordering process for a restaurant company operating multiple branches. The system will allow customers to place pickup and delivery orders, while restaurant staff manage daily business operations through role-based access.

The core business functions included within the project scope are:

- Branch Management
- Menu Management
- Customer Ordering
- Order Fulfillment Management
- Delivery Management
- Complaint and Review Management

Supporting functionalities such as user authentication, user account management, notifications, and payment processing will also be included to support the operation of the system.

Project Limitations

The proposed system has the following limitations:

- The system is designed for a single restaurant company operating multiple branches.
- The application is web-based and does not include a native mobile application.
- Delivery services are limited to predefined delivery areas configured by each branch.
- Real-time GPS navigation and live delivery tracking are not included.
- Integration with third-party food delivery platforms is outside the scope of this project.

3. System Overview

System Overview Diagram

4. Functional Requirements

1. Branch Management

The system shall support the management of restaurant branches, including branch information, operating hours, delivery areas, branch manager assignments, and branch performance monitoring.

2. Menu Management

The system shall support the management of food categories, menu items, item variations, pricing, availability, and food images for each restaurant branch.

3. Customer Ordering

The system shall allow customers to browse branch-specific menus, place pickup or delivery orders, manage delivery addresses, manage shopping carts, track order status, and view order history.

4. Order Fulfillment Management

The system shall support order confirmation, order preparation, order status updates, preparation time estimation, rider assignment, and order completion for pickup and delivery orders.

5. Delivery Management

The system shall support delivery rider availability management, delivery assignment, delivery status updates, and delivery history management.

6. Complaint and Review Management

The system shall allow customers to submit complaints and reviews while enabling customer service staff to manage complaint resolution and monitor customer feedback.

Supporting Functionalities

In addition to the core business functions, the system shall support user authentication, user account management, role-based access control, payment processing, notifications, and profile management to ensure the effective operation of the system.

5. Non-Functional Requirements

Performance: The system shall respond to user requests within a reasonable time under normal operating conditions and efficiently support multiple users simultaneously.

Security: The system shall require user authentication and role-based access control to ensure that users can only access functions relevant to their assigned roles. User passwords shall be securely stored.

Usability: The system shall provide a user-friendly and intuitive interface with clear and consistent navigation, allowing users to perform tasks with minimal training.

Availability: The system shall be available whenever required, except during scheduled maintenance, while minimizing system downtime.

Scalability: The system shall support future growth, including additional branches, users, menu items, and customer orders, without requiring major system modifications.

Reliability: The system shall maintain accurate and consistent data while ensuring that customer orders and business information are processed correctly.

Maintainability: The system shall be developed using a modular structure to simplify future enhancements, maintenance, and bug fixes.

Compatibility: The system shall be accessible through modern web browsers on desktop and mobile devices.

6. Major Stakeholders / Users

The proposed **Spice Avenue** system will be used by several stakeholders, each with specific responsibilities to support the restaurant's daily operations and provide an efficient food ordering experience.

User Role	Description
Customer	Places pickup or delivery orders, manages delivery addresses, tracks order status, submits complaints and reviews, and manages their account.
Branch Manager	Manages menus, food categories, item variations, pricing, availability, operating hours, delivery areas, and processes customer orders within the assigned branch.
Delivery Rider	Manages rider availability, accepts assigned deliveries, updates delivery statuses, and completes customer deliveries.
Customer Service Supervisor	Handles customer complaints, updates complaint statuses, resolves customer issues, and monitors customer reviews and feedback.
Operations Manager	Manages restaurant branches, assigns Branch Managers, monitors branch performance, and oversees branch operations.
System Administrator	Manages staff user accounts, assigns user roles, resets passwords, and maintains system security and access control.

7. Six Major Functions

Member 1 – Branch Management

Who Will Use It

- Operations Manager
- Branch Manager

What It Does

Manages restaurant branches, including branch information, manager assignments, operating hours, delivery areas, and branch performance.

Expected Outcome

Branches operate efficiently with accurate information and effective service coverage.

Member 2 – Menu Management

Who Will Use It

- Branch Manager

What It Does

Manages food categories, menu items, item variations, pricing, availability, and food images.

Expected Outcome

Customers have access to accurate and up-to-date branch menus.

Member 3 – Customer Ordering

Who Will Use It

- Customer

What It Does

Allows customers to place pickup or delivery orders, manage delivery addresses, browse menus, manage shopping carts, and track orders.

Expected Outcome

Customers can order food conveniently through a simple online ordering process.

Member 4 – Order Fulfillment Management

Who Will Use It

- Branch Manager

What It Does

Processes customer orders by confirming orders, updating order statuses, setting preparation times, and assigning delivery riders.

Expected Outcome

Orders are processed efficiently from confirmation to pickup or delivery.

Member 5 – Delivery Management

Who Will Use It

- Delivery Rider

What It Does

Manages delivery assignments, rider availability, delivery status updates, and completed deliveries.

Expected Outcome

Delivery orders are completed efficiently with accurate status updates.

Member 6 – Complaint and Review Management

Who Will Use It

- Customer
- Customer Service Supervisor

What It Does

Manages customer complaints, reviews, and feedback to support service improvement.

Expected Outcome

Customer issues are resolved effectively, and feedback helps improve service quality.

8. Minor Functions

In addition to the six major business functions, the proposed **Spice Avenue** system will provide the following supporting functions to ensure secure and efficient system operation:

- **User Registration:** Allow new customers to create an account.
- **User Login:** Authenticate registered users before accessing the system.
- **User Logout:** Allow users to securely end their sessions.
- **Password Reset:** Allow users to reset forgotten passwords.
- **User Profile Management:** Allow users to view and update their profile information.
- **Role-Based Access Control:** Restrict system functions based on user roles.
- **Payment Processing:** Support card payment and Cash on Delivery (COD) during checkout.
- **System Notifications:** Notify users about order confirmations, status updates, and complaint resolutions.
- **Search Functionality:** Allow users to search menu items by name or category.

9. System Limitations / Constraints

The proposed **Spice Avenue** system is subject to the following limitations and constraints:

- The system is designed as a **web-based application** and requires a stable internet connection to access its features.
- The system is intended to support **modern web browsers** on desktop and mobile devices. Compatibility with older browsers is not guaranteed.
- Delivery service is limited to the **delivery areas configured by each Branch Manager**. Customers outside these areas will not be able to place delivery orders from that branch.
- The system does **not provide real-time GPS tracking or navigation** for delivery riders. Delivery status updates are managed manually by riders.
- Payment is limited to **Card Payment** and **Cash on Delivery (COD)**. Integration with external payment gateways is outside the scope of this project.
- The system assumes that restaurant staff will maintain accurate information, including menu items, pricing, operating hours, delivery areas, and order status updates.
- Notifications are limited to **system-generated status updates within the application** and do not include SMS or email notifications.
- The system is designed for a **single restaurant business operating multiple branches** and is not intended to support multiple restaurant companies.

10. Simple Timeline

Week 3

Understand the project requirements, identify system users, define the project scope, and finalize the six major functions.

Week 4

Analyze the system requirements and prepare the overall system design, including workflows and business rules.

Week 5

Design the database structure and user interface prototypes.

Week 6

Set up the development environment and begin implementing the database and core system structure.

Week 7

Develop the Branch Management and Menu Management functions.

Week 8

Develop the Customer Ordering function.

Week 9

Develop the Order Fulfillment Management function and integrate the completed modules.

Week 10

Develop the Delivery Management and Complaint & Review Management functions.

Week 11

Complete the remaining system features and integrate all modules.

Week 12

Perform system testing and fix identified issues.

Week 13

Refine the user interface, improve system performance, and complete project documentation.

Week 14

Conduct final testing, fix remaining issues, and finalize the system.

11. Conclusion

The proposed **Spice Avenue** is a web-based food ordering system designed to improve the efficiency of restaurant operations across multiple branches while providing customers with a convenient and user-friendly ordering experience. The system integrates branch management, menu management, customer ordering, order fulfillment, delivery management, and complaint and review management into a single centralized platform. This enables restaurant staff to manage daily operations more effectively while allowing customers to place and track orders with ease.

The implementation of this system is expected to reduce manual processes, improve communication between customers and restaurant staff, enhance order accuracy, and support better decision-making through branch performance monitoring. By providing both pickup and delivery services, the system also offers greater flexibility and convenience to customers.

To successfully complete the project, the development work will be divided among the six group members, with each member responsible for one major system function. The team will follow a structured development approach, including requirements analysis, system design, implementation, testing, and documentation. Regular communication, collaboration, and progress reviews will ensure that all components are integrated successfully and the project objectives are achieved within the planned timeline.

Overall, the proposed system provides a practical and scalable solution for managing food ordering operations and is expected to deliver significant benefits to both customers and restaurant management.